

FIT-BUDDY AI – PROJECT DEVELOPMENT PHASE

1. Set Up Development Environment

- Install required software and development tools.
- Create the project structure.
- Configure the Gemini API and database.

2. Develop User Interface

- Create login and registration pages.
- Develop the user profile section.
- Create fitness goal and preference forms.
- Design the workout and progress dashboards.

3. Develop Backend

- Implement server-side logic.
- Connect the frontend with the backend.
- Manage user accounts and fitness data.

4. Integrate Gemini AI

- Connect the Gemini API.
- Send user requirements to Gemini.
- Generate personalized workout plans.
- Implement AI-based plan modification.

5. Develop Workout Module

- Display daily workout plans.
- Show exercise details, duration, and repetitions.
- Allow users to mark workouts as completed.

6. Develop AI Chatbot

- Allow users to ask fitness-related questions.
- Generate responses using Gemini.
- Handle user feedback and requests for workout changes.

7. Develop Progress Tracking

- Store completed workout information.
- Maintain workout history.
- Display user progress and consistency.

8. Implement Database

- Store user profiles.
- Store workout plans and history.
- Store feedback and progress data.

9. Implement Security and Safety

- Secure user authentication.
- Protect stored user information.
- Add safety checks for AI-generated fitness recommendations.

10. Module Integration

- Connect UI, backend, database, and Gemini AI.
- Ensure all modules communicate correctly.

11. Testing During Development

- Test individual modules.
- Check Gemini responses.
- Fix bugs and integration issues.
- Test different user scenarios.

12. Final Development

- Optimize the application.
- Improve UI and performance.
- Prepare the completed Fit-Buddy AI system for final testing and deployment.